

ACJC 2018 JC1 H2 Chemistry Promo Exam Solutions**Section A (15 marks): SOLUTIONS**

1	2	3	4	5	6	7	8	9	10
C	D	C	B	C	B	A	D	A	B
11	12	13	14	15					
A	B	D	A	B					

Solution for Q1

$$\text{Angle of Deflection} \propto \frac{\text{Charge}}{\text{Mass}}$$

Charge / Mass

- Option A: $+1 / 7 = +0.143$
 Option B: $-2 / 16 = -0.125$
 Option C: $+2 / 24 = +0.083$ (smallest magnitude)
 Option D: $-3 / 31 = -0.097$

Solution for Q2

Amount of $\text{BrO}_3^- = (0.015)(0.02) = 0.0003 \text{ mol}$
 Amount of electrons taken in by $\text{BrO}_3^- = (6)(0.0003) = 0.0018 \text{ mol}$

Amount of electrons given out by $\text{NH}_2\text{OH} = 0.0018 \text{ mol}$
 Amount of $\text{NH}_2\text{OH} = (0.040)(0.015) = 0.0006 \text{ mol}$

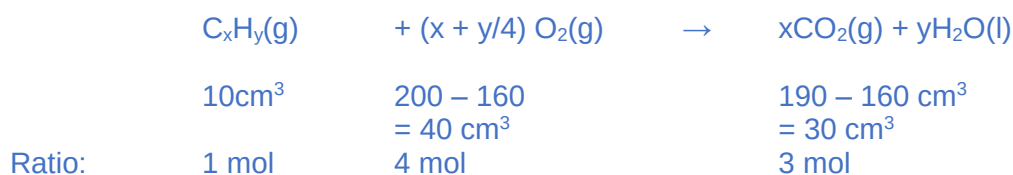
Solution for Q3

Option A: Collisions between the gas particles should be elastic.

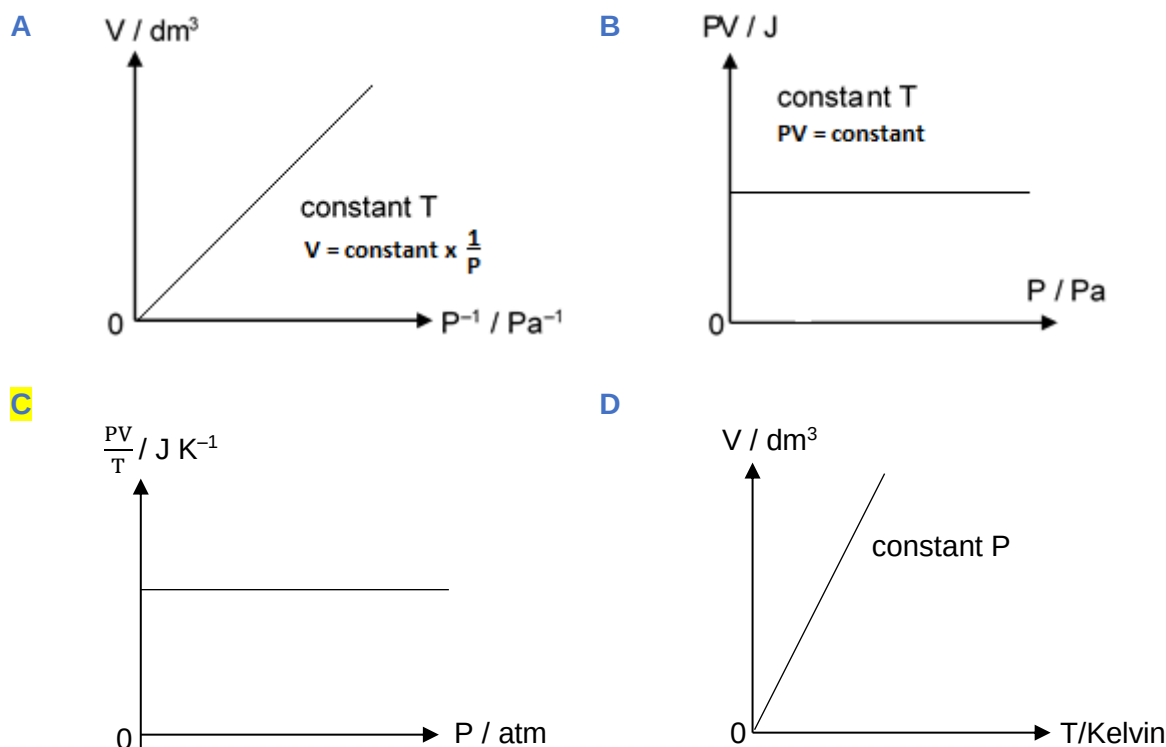
Option B: The volume of gas particles is negligible compared to volume of gas.

Option C: Correct.

Option D: The average kinetic energy of a gas particle is proportional to the temperature (in terms of Kelvin) of the gas.

Solution for Q4

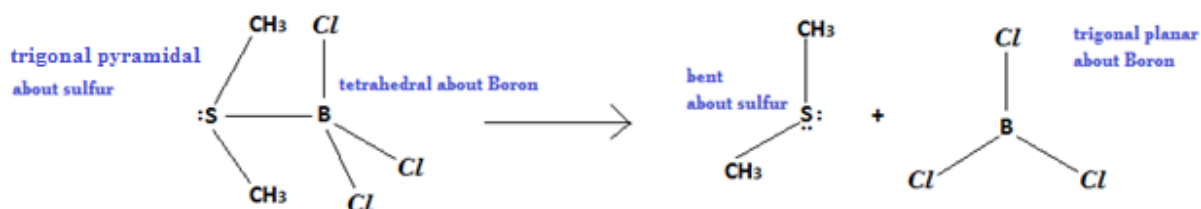
Solution for Q5



Solution for Q6

- A** All the bond angles surrounding each oxygen atom in ICE are 109.5°.
- B** Four electrons from each oxygen are involved in forming hydrogen bonds.
- C** The hydrogen bonds, shown by the dotted lines, are weaker than the O-H covalent bonds.
- D** The open structure of ice causes ice to be less dense than water.

Solution for Q7



- A** The C-S-C bond angle decreases when the solid decomposes.
- B** The dative bond is from sulfur to boron.
- C** The dative bond is formed using the hybrid sp^3 orbitals of boron and sulfur.
- D** $(CH_3)_2S$ and BCl_3 act as the Lewis base and Lewis acid respectively in the formation of $(CH_3)_2S.BCl_3$.

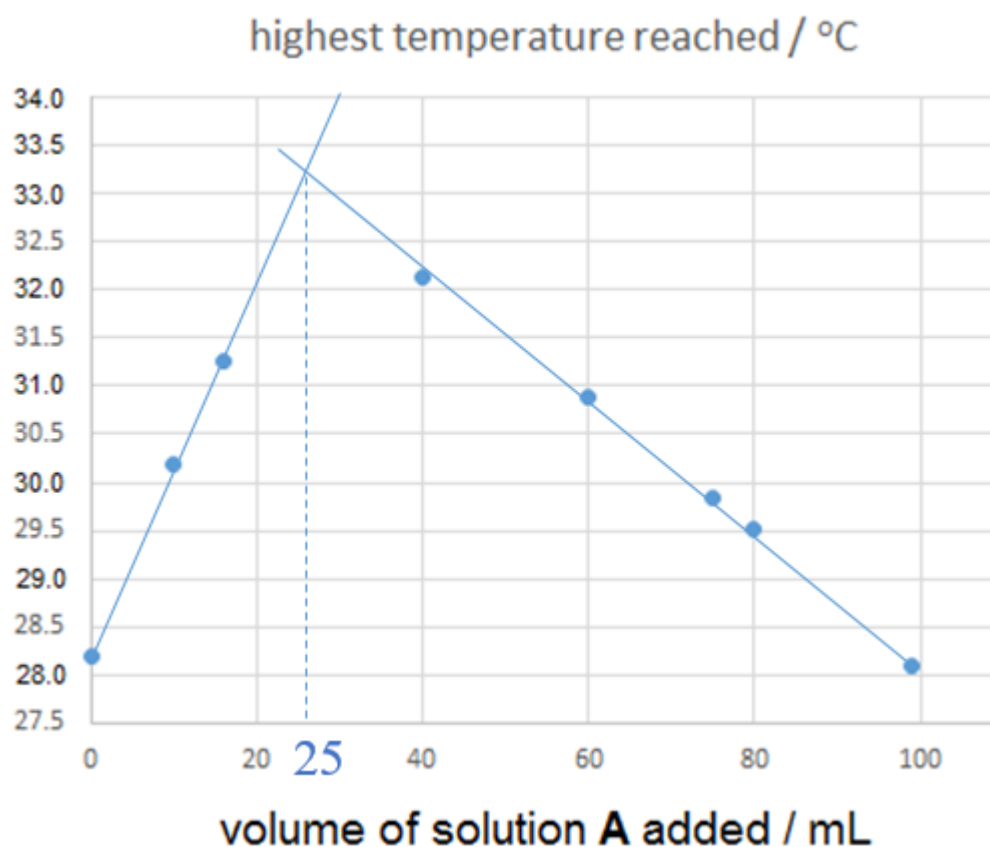
Solution for Q8

All ions are isoelectronic, but different number of protons.

	Sr^{2+}	Rb^{+}	Se^{2-}	As^{3-}
Proton / Electron Ratio	$38 \div 36$	$37 \div 36$	$34 \div 36$	$33 \div 36$
Attraction	Strongest			Weakest

Solution for Q9

- 1 E_3 is the activation energy of the second step.
- 2 ΔH can be obtained by subtracting E_3 from E_1 .
- 3 There may not be equal number of intermediates and transition states.

Solution for Q10

25 cm³ of A react with 75 cm³ of B.

Solution for Q11

[Reactant] = 0.48 → 0.24 → 0.12 → 0.06
 Half-Life 14 min 14 min 14 min

[Reactant] = 0.40 → 0.20 → 0.10
 Half-Life 14 min 14 min

Solution for Q12

Compare Expt 1 and 3
 Keeping [A] constant, and Double [B], Rate Double
 Order of Reaction wrt B is ONE.

Compare Expt 2 and 3
 Keeping [B] constant, and Double [A], Rate remains the same.
 Order of Reaction wrt A is ZERO.

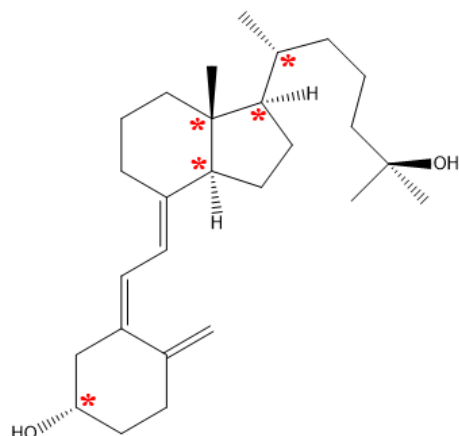
Hence A is not part of the rate-determining step.
 Using Expt 1 data, Rate = $k [B]$ Therefore, $k = 1$
 Unit of rate constant is $(\text{time})^{-1}$

Solution for Q13

- A** After reaction, salt and strong acid are present. Not buffer.
- B** Weak Base and Strong Alkali present. Not buffer.
- C** Salt only. Not buffer.
- D** After reaction, salt and weak acid are present. This is acid buffer.

Solution for Q14

- A** At higher $[H^+]$
 $H^+(aq) + CO_3^{2-}(aq) \rightarrow HCO_3^-(aq)$
 $CdCO_3(s) \rightleftharpoons Cd^{2+}(aq) + CO_3^{2-}$
 $[CO_3^{2-}]$ decreases, and EQM shifts to the right. Hence solubility increases.
 This is similar for the other three salts.
 $H^+(aq) + S^{2-}(aq) \rightarrow HS^-(aq)$
- B** Units are different. We need to calculate solubility from K_{sp} values.
 CuS is less soluble than Ag_2S .
- C** Units are the same. Hence we can compare the K_{sp} values.
 $CoCO_3$ is more soluble than $CdCO_3$.
- D** The K_{sp} value remains constant if temperature remains constant.
 However, the solubility of CuS will decrease in this case.

Solution for Q15

- 1 There are FIVE chiral carbons.
- 2 There are 21 sp^3 hybridised carbon atoms.
Check for tetrahedral carbon atoms.
- 3 TWO $\text{C}=\text{C}$ bonds show cis-trans isomerism.
ONE $\text{C}=\text{C}$ bond does not due to two H atoms on the same C in $\text{C}=\text{C}$ bond.

END OF SECTION A

Section B (35 marks)

Answer **all** questions in the spaces provided.

1 Uranium, U, is an important element in nuclear research.

(a) Uranium-235 (^{235}U) is an isotope of uranium making up less than one percent of natural uranium.

(i) State the number of neutrons and protons in a ^{235}U atom. [1]

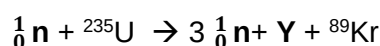
143 neutrons. 92 protons.

(ii) All the isotopes of uranium are unstable.

When a ^{235}U atom is bombarded with a neutron, three neutrons, an atom of krypton-89 and a particle, **Y** are formed.

Y contains 88 neutrons and other particles.

Schematically,



Determine the identity of **Y**. [1]

Y: barium-144

(b) The most stable isotope of uranium is uranium-238.

For **(b)**, you are to use 238.0 as the relative atomic mass of uranium.

Excess water was added to 4.267 g of UF_6 .

The only products were 3.730 g of a solid **Z** containing only uranium, oxygen and fluorine and 0.970 g of HF gas. The molecular formula of **Z** contains one U atom.

(i) Calculate the number of moles of HF and UF_6 . [2]

$n(\text{UF}_6) = 1.21 \times 10^{-2}$ moles

$n(\text{HF}) = 4.85 \times 10^{-2}$ moles

- (ii) Hence, state the number of moles of **Z** formed. [1]

$$n(\mathbf{Z}) = 1.21 \times 10^{-2} \text{ moles}$$

- (iii) Using your answer in **b(ii)**, determine the molar mass of **Z**. [1]

$$\text{Molar mass of } \mathbf{Z} = 307.8 \text{ g mol}^{-1}$$

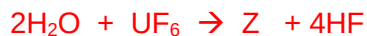
- (iv) Using your answers in **(b)**, determine the molecular formula of **Z**. [3]

$$\text{Amt of UF}_6 = 0.0121 \text{ mol}$$

$$\text{Amt of HF} = 0.0485 \text{ mol}$$

$$\text{UF}_6 : \text{HF}$$

$$1 \text{ mole} : 4 \text{ moles}$$



$$\text{No. of moles of F per mole of } \mathbf{Z} = 6 - 4 = 2$$

$$\text{Mass of oxygen per mole of } \mathbf{Z} = 307.8 - 2(19) - 238 = 31.8$$

$$\text{No. of moles of O per mole of } \mathbf{Z} = 2$$

$$\text{Molecular formula of } \mathbf{Z} = \text{UO}_2\text{F}_2$$

[Total: 9]

- 2 Disulfur decafluoride (S_2F_{10}) is an extremely toxic gas which almost became a chemical weapon during World War II.

It exists in equilibrium with sulfur tetrafluoride (SF_4) and sulfur hexafluoride (SF_6).



- (a) There is a S-S single bond in disulfur decafluoride. All the ten fluorine atoms are in equivalent environments in disulfur decafluoride.

- (i) State the shape with respect to each sulfur in S_2F_{10} . [1]

octahedral / square bipyramidal wrt each S.

- (ii) The boiling points of the three compounds are tabulated below.

Compound	SF ₄	SF ₆	S ₂ F ₁₀
Boiling point / K	235	209	303

Explain why SF₆ has a lower boiling point than SF₄.

[2]

SF₆ is octahedral with 6 equal bonds hence it is non-polar.

SF₄ is see-saw with 4 equal bonds hence it is polar.

As permanent dipole – permanent dipole (pd-pd) interactions are generally stronger than instantaneous dipole-induced dipole (id-id) interactions, SF₆ has the lower bp.

- (iii) Explain why SF₄ has a lower boiling point than S₂F₁₀.

[1]

S₂F₁₀ is a larger molecule and hence the electron cloud is larger and more polarisable. Stronger instantaneous dipole-induced dipole (id-id) interactions exist between its molecules than between SF₄ molecules and therefore higher boiling point.

- (b) This part of the question focuses on the thermodynamic aspects of the reaction.



- (i) Using suitable data from the Data Booklet, determine the enthalpy change of the reaction.

State the most important assumption you made.

[2]

$$\Delta H = \text{BE}(\text{S-S}) = + 264 \text{ kJ mol}^{-1}$$

The S-F bonds are all equal in S₂F₁₀, SF₄ and SF₆.

Or Each S-F bond has a different BE but in this case assuming it is the same.

- (ii) Explain why this decomposition becomes more thermodynamically feasible at higher temperatures. [2]

ΔS is positive

due to the increase in amount of gas particles.

$-T\Delta S$ is negative

T has to be as high as possible to make ΔG as negative as possible / offset the positive ΔH .

- (c) This part of the question focuses on the chemical equilibrium aspects of the reaction.



In a study of this decomposition, 2 moles of disulfur decafluoride was placed in a 2.00 dm³ flask and heated up to 100 °C; its concentration became 0.50 mol dm⁻³ at equilibrium.

- (i) Calculate the value of K_c at 100 °C. Include the units. [2]

ICE Table

$$K_c = 0.50 \text{ mol dm}^{-3}$$

To this equilibrium mixture, a certain mass of disulfur decafluoride was added. The new equilibrium concentration of it became 2.50 mol dm⁻³.

- (ii) Calculate the mass of disulfur decafluoride added. [3]

$$\begin{aligned} &\text{Decrease in concentration of } \text{S}_2\text{F}_{10} \\ &= (2.50 / 2)^{0.5} - 0.5 = 0.618 \text{ mol dm}^{-3} \end{aligned}$$

$$x = 2.50 + y - 0.50 = 2.50 + 0.618 - 0.50 = 2.62 \text{ mol dm}^{-3}$$

$$n(\text{S}_2\text{F}_{10}) \text{ added} = 2 \text{ dm}^3 \times (2.62) \text{ mol dm}^{-3} = 5.24 \text{ mol}$$

$$\text{Mass of } \text{S}_2\text{F}_{10} \text{ added} = 5.24 \text{ moles} \times 254 \text{ g mol}^{-1} = 1.33 \text{ kg}$$

- (iii) Predict, with reasoning, how the *position of equilibrium* would change when the volume of the container is reduced. [1]

Pressure increases hence favour the direction that decreases the number of gaseous particles which in this case is the backward reaction, hence POE shifts left.

[Total: 14]

- 3 This question is about interhalogens, compounds that are made of more than one type of halogen.

Chlorine trifluoride, ClF_3 , is one of the most reactive chemicals ever known to man. It causes sand to explode.

- (a) ClF_3 reacts with silver chloride according to the **unbalanced** equation below.



- (i) Balance the above equation by inserting the respective stoichiometric coefficients in the spaces provided. Do fill in the coefficient even if it happens to be 1. [1]



- (ii) State the electronic configuration of the outermost shell of Ag in AgCl . [1]

$4s^2 4p^6 4d^{10}$

- (b) ClF_3 and ClBr_3 share the same molecular shape. However, the F-Cl-F bond angle is predicted to be smaller than the Br-Cl-Br angle.

- (i) Explain why bromine is less electronegative than fluorine. [1]

Electrons getting further away from the nucleus. Therefore, nuclear attraction gets weaker for the outermost electrons as distance increases.

- (ii) Hence, or otherwise, explain the relative sizes of the two bond angles. [2]

The electron density of the bond pairs are closer to the central atom in C/Br_3 than in C/F_3 .

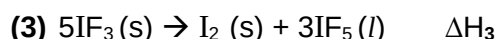
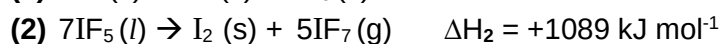
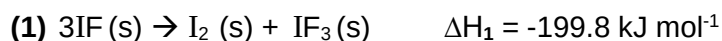
The bond pair-bond pair repulsion in C/Br_3 is stronger.

OR

The F atom is smaller than the Br atom, more steric repulsion in C/Br_3

- (c) Iodine forms many compounds with fluorine – IF , IF_3 , IF_5 and IF_7 .

The four compounds can interconvert into one another according to the following chemical equations.



- (i) All three reactions can be classified under a certain type of redox reaction. Identify this type of reaction. [1]

Disproportionation

The standard enthalpy changes of formation of IF and IF_7 are -95.4 and $-962.5 \text{ kJ mol}^{-1}$ respectively.

- (ii) Calculate the standard enthalpy changes of formation of IF_3 and that of IF_5 . [2]

$$-199.8 = \Delta H_f(\text{IF}_3) - 3(-95.4) \rightarrow \Delta H_f(\text{IF}_3) = -486 \text{ kJ mol}^{-1}$$

$$1089 = 5(-962.5) - 7\Delta H_f(\text{IF}_5) \rightarrow \Delta H_f(\text{IF}_5) = -843 \text{ kJ mol}^{-1}$$

- (iii) Hence, calculate the standard enthalpy change of reaction (3). [1]

$$3\Delta H_f(\text{IF}_5) - 5\Delta H_f(\text{IF}_3) = 3(-843) - 5(-486) = -99 \text{ kJ mol}^{-1}$$

- (iv) Briefly explain why there are negligible changes in entropy in reactions (1) and (3). [1]

In these two reactions there is no change in the amount of gas.

Or

Rxn 1 all solids not much change in randomness and rxn 3 change from solid to liquid and hence degree of randomness does not increase very much.

- (v) Hence, predict whether reaction (1) or (3) has a lower likelihood of happening. [2]

The change in entropy (or $T\Delta S$) for the two reactions may be assumed to be equal. $T\Delta S$ may be ignored.

Reaction (3) is less exothermic

Hence ΔG for reaction (3) is less negative hence reaction (3) is less spontaneous.

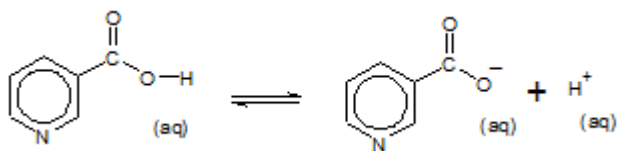
[Total: 12]

~ END OF SECTION B ~

Section C (25 marks)

Answer **all** questions on foolscap paper.

- 1 Nicotinic acid is a form of vitamin B3, an essential human nutrient. Insufficient nicotinic acid in the diet can cause nausea, anemia, headaches, and tiredness.



Nicotinic Acid

It is given that the K_a value for nicotinic acid at 25°C is 1.4×10^{-5} .

- (i) A student pipettes out 25.0 cm³ of 0.1 mol dm⁻³ nicotinic acid (aq) and transfers it into a conical flask. Calculate the pH of this solution. **[1]**

	HA (aq)	$\rightleftharpoons$	A^- (aq) +	H^+ (aq)
Initial / mol dm ⁻³	0.1		0	0
Change	- x		+ x	+ x
Eqm	0.1 - x		x	x

$$x^2 \div (0.1 - x) = 1.4 \times 10^{-5}$$

Assume that x is small, ie. $(0.1 - x) \approx 0.1$

$$x = 1.18 \times 10^{-3}$$

$$\text{pH} = -\log(1.18 \times 10^{-3}) = 2.93$$

- (ii) He then titrates the solution with 0.2 mol dm⁻³ NaOH (aq) using a suitable indicator. Calculate the pH of solution when endpoint is reached. **[3]**

$$\text{Initial Amount of Nicotinic Acid} = (0.025)(0.1) = 0.0025 \text{ mol}$$

At endpoint,

$$\text{Amount of Nicotinic Acid} = 0.0025 \text{ mol}$$

$$\text{Volume of NaOH used} = 0.0025 \div 0.2 = 0.0125 \text{ dm}^3 = 12.5 \text{ cm}^3$$

$$[A^-] = 0.0025 \div (25.0 + 12.5) = 0.0667 \text{ mol dm}^{-3}$$

	A^- (aq) +	H_2O (l)	$\rightleftharpoons$	HA (aq) +	OH^- (aq)
Initial / mol dm ⁻³	0.0667			0	0
Change	- x			+ x	+ x
Eqm	0.0667 - x			x	x

$$K_b = K_w \div K_a = 7.14 \times 10^{-10} \text{ mol dm}^{-3}$$

$$x^2 \div (0.0667 - x) = 7.14 \times 10^{-10}$$

Assume that x is small, ie. $(0.0667 - x) \approx 0.0667$

$$x = 6.90 \times 10^{-6}$$

$$\text{pOH} = -\log(6.90 \times 10^{-6}) = 5.16$$

$$\text{pH} = 14 - \text{pOH} = 8.84$$

- (iii) Calculate the pH at maximum buffer capacity. [1]

$$\text{pH} = \text{pK}_a = 4.85$$

- (iv) Calculate the maximum pH of solution when NaOH(aq) is added in excess. [1]

$$\text{pOH} = -\log (0.2) = 0.7$$

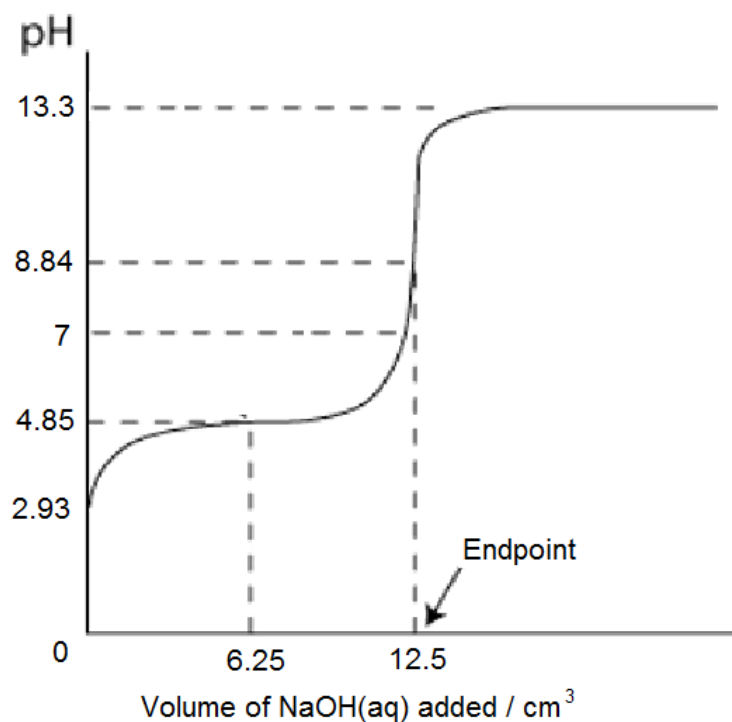
$$\text{pH} = 14 - \text{pOH} = 13.3$$

- (v) Suggest a suitable indicator and state its colour change at endpoint. [1]

Phenolphthalein

Endpoint colour change: From colourless to pink.

- (vi) Sketch the titration curve. Indicate all points calculated from (a)(i) to (iv) on the curve. [1]



[Total: 8]

- 2 The milk of magnesia is a short-term medication used to treat constipation, heartburn and indigestion. Its active ingredient is magnesium hydroxide, $\text{Mg}(\text{OH})_2$.

$\text{Mg}(\text{OH})_2$ is a sparingly soluble salt with a solubility product of $5.61 \times 10^{-12} \text{ mol}^3 \text{ dm}^{-9}$ at 25 °C.

- (a) Calculate the solubility of $\text{Mg}(\text{OH})_2$ in water at 25 °C. [1]

$$4x^3 = 5.61 \times 10^{-12} \rightarrow$$

$$x = 1.12 \times 10^{-4} \text{ mol dm}^{-3}$$

- (b) Calculate the solubility of $\text{Mg}(\text{OH})_2$ in a 0.2 mol dm^{-3} magnesium nitrate solution at 25 °C. [2]

$$(0.2 + x)(2x)^2 = 5.61 \times 10^{-12}$$

$$\text{Solving, } x = 2.65 \times 10^{-6} \text{ mol dm}^{-3}$$

- (c) Calculate the minimum pH needed to precipitate out $\text{Mg}(\text{OH})_2$ from a magnesium nitrate solution to the extent that the resulting $[\text{Mg}^{2+}]$ is no higher than $1 \times 10^{-7} \text{ mol dm}^{-3}$ at 25 °C. [2]

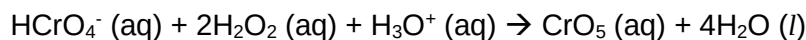
$$[\text{OH}^-] = \sqrt{\frac{5.61 \times 10^{-12}}{10^{-7}}} = 7.49 \times 10^{-3} \text{ mol dm}^{-3}$$

$$\text{pH} = 14 + \log (7.49 \times 10^{-3}) = 11.9$$

[Total: 5]

- 3** Chromium(VI) peroxide or chromium oxide peroxide (CrO_5) is an even more powerful oxidising agent than potassium dichromate(VI).

It is a highly unstable compound formed by the addition of acidified hydrogen peroxide solutions to solutions of metal chromates or dichromates, such as sodium chromate or potassium dichromate.



The generally yellow chromates or orange dichromates turn to dark blue as chromium(VI) peroxide is formed.

- (a)** Explain, why the oxidation state of chromium cannot be +10, as what the molecular formula CrO_5 suggests. [1]

The maximum oxidation state is +6 / it only has six valence electrons.

Or

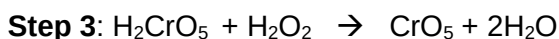
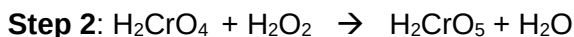
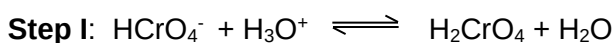
Removing the four additional electrons from the inner principal quantum shell requires a lot of energy

- (b)** The kinetics of this reaction have been studied, and the rate equation is found to be

$$\text{Rate} = k[\text{HCrO}_4^-][\text{H}_3\text{O}^+][\text{H}_2\text{O}_2]$$

- (i)** State the units of k , the rate constant. $\text{mol}^{-2} \text{dm}^6 \text{s}^{-1}$ [1]

One of the proposed mechanisms for this reaction is as follows



- (ii)** Suggest a certain physical property that can be monitored across time, in order to follow the progress of the above reaction. [2]

Electrical conductivity or pH

decreases / Increases across time (respectively)

- (iii)** Show that step 2 is the rate determining step. [2]

From step 1, $[\text{H}_2\text{CrO}_4] = K_c[\text{HCrO}_4^-][\text{H}_3\text{O}^+]$

$$\text{Rate} = k[\text{H}_2\text{CrO}_4][\text{H}_2\text{O}_2]$$

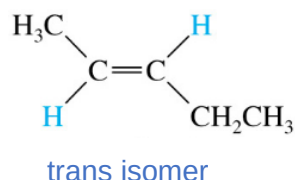
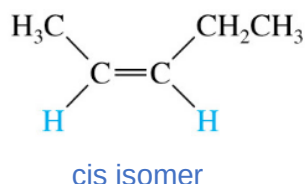
$$= k(K_c[\text{HCrO}_4^-][\text{H}_3\text{O}^+])[\text{H}_2\text{O}_2]$$

$$= k'[\text{HCrO}_4^-][\text{H}_3\text{O}^+][\text{H}_2\text{O}_2]$$

[Total: 6]

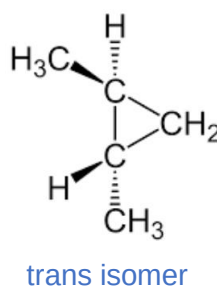
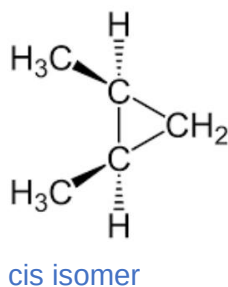
- 4 Pentenes are components of gasoline and are often produced as by-products of thermal cracking of petroleum. Pent-2-ene is one of the isomers of pentene. It has the following formula: $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3$.

(a) Draw and label the two structures of pent-2-ene that show cis-trans isomerism. [2]



One constitutional isomer of pent-2-ene is the 1,2-dimethylcyclopropane.

(b)(i) Draw and label the two structures of 1,2-dimethylcyclopropane that show cis-trans isomerism. [2]



(ii) For each of the two structures drawn in (b)(i), explain whether it is optically active. [2]

Though the cis isomer has two chiral carbons, it has a plane of symmetry. Molecule is not chiral. It is a meso compound. Sample is optically not active.

The trans isomer has two chiral carbons and no plane of symmetry. Molecule is chiral. Sample is optically active.

[Total: 6]

END OF PAPER